

The utilization of artificial intelligence in improvising the rural health infrastructure of India

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The utilization of artificial intelligence in improvising the rural health infrastructure of India

Abstract

Over the years, India has been proactively pursuing improvements in healthcare infrastructure and capacity building. The country is constantly striving towards improving its disparity of healthcare services in the rural and the urban centres. Although there has been research on rural health infrastructure in the past, there remains a dearth of models which can predict to an extent the number of sub centres available in a particular state. The objective of the model is to predict the number of sub centres in various states of India. The research consisted of the development of three machine learning models, a logistic regression model, a random forest classifier and a neural network model. We hypothesized that our methods would work sufficiently providing data for the number of sub centres in a particular state in India. A dataset sourced from Kaggle¹ was used in respective codes to train the model. The percentage difference in the training and the testing dataset for the neural network model was as low as 0.005%, while for the linear regression model it was 0.04% and for the random forest classifier it was 0.05%. The model can be used for any development plan, wherein resources required can be estimated for the already existing structures, through an estimate of the number of centres already in place.

Introduction

Health Infrastructure consists of doctors, nurses, hospitals, and various paramedical professionals, equipment and beds needed in the hospitals and a well-developed pharmaceutical industry. By developing health infrastructure, a country can ensure healthy manpower for the production of goods and services. It is possible only when the health infrastructure is available to all. Since independence, there has been significant development in physical healthcare services and improvements in health indicators, but this is insufficient for India's rapidly increasing population. There is a significant disparity in healthcare utilisation between rural and urban regions as well as between the poor and the rich. With reports of increased incidents of female foeticide and death, the health, especially of women, across the country has become a major issue.

Despite the fact that 70% of India's population lives in rural areas, only one-fifth of its hospitals (including private hospitals) are located in rural areas.² Rural India has around half the number of dispensaries. As a result, residents in rural regions lack sufficient healthcare infrastructure.

Nevertheless, over the last five decades, India has made considerable progress in the improvement of the health of the people by building up a good health infrastructure. The government has set up various hospitals (known as Primary Health Centres or PHCs), at the village level. Primary Health Centres are state-owned public health facilities that provide health care services in the villages. Besides, a large

¹ <https://www.kaggle.com/datasets/vineethakkinapalli/rural-health-statistics-india>

² *A study on the health infrastructure of India, its development over the years and current status on*
<https://www.geeksforgeeks.org/health-infrastructure-in-india/>

number of hospitals are run by voluntary agencies and the private sector. Community health centres are care facilities with the goal of increasing access to crucial primary and preventative care services.³ CHCs primarily serve patients from lower income areas who may be uninsured or underinsured, lack access to transportation, or face language barriers. A typical Anganwadi centre provides basic health care in a village. It is a part of the Indian public health care system.⁴ Professionals and Para-medical Professionals who are trained in pharmacy, medical, and nursing colleges control these centres.

Although there has been research on rural health infrastructure in the past, there remains a dearth of models which can predict to an extent the number of sub centres available in a particular state. The objective of the model is to predict the number of sub centres in various states of India, given the number of PHC (Primary Healthcare Centre), CHC (Community Health Centres), ANM (Auxiliary Nursing Midwifery), Pharmacists and Nursing Staff. Given the developing stage of India's rural infrastructure, it is crucial to have an estimate of the number of structures already in place, to further estimate how many more are required in proportion to the population of that area. Additionally, for any development plan, resources required can be estimated for the already existing structures, through an estimate of the number of models already in place.

Materials and methods:

Libraries used:

The model was based around Pandas and Scikit-learn. The Python Standard Library contains the exact syntax, semantics, and tokens of Python. It contains built-in modules that provide access to basic system functionality like I/O and some other core modules.⁵

Python libraries play a very vital role in fields of Machine Learning, Data Science, Data Visualization, etc. Pandas provided flexible high-level data structures and a variety of analysis tools. It eased data analysis, data manipulation, and cleaning of data. Pandas supported operations like Sorting, Re-indexing, Iteration, Concatenation, Conversion of data, Visualizations, Aggregations, etc. It allowed for simple data modelling and data analysis operations without the need to switch to another language such as R.

Scikit-learn supported variously supervised and unsupervised algorithms like linear regression, classification, clustering, etc. This library works in association with NumPy and SciPy. The name "NumPy" stands for "Numerical Python". It is the commonly used library that supports large matrices and multi-dimensional data for this model. It consists of in-built mathematical functions for easy computations. The name "SciPy" stands for "Scientific Python". This library is built over an extension of NumPy. It

³ *Introduction to the community health centres in India*,
<https://www.definitivehc.com/resources/glossary/community-health-center>

⁴ *In depth study of Anganwadi in the Indian state*
[https://en.wikipedia.org/wiki/Anganwadi#:~:text=July%202020\),%22courtyard%20shelter%22%20in%20English.](https://en.wikipedia.org/wiki/Anganwadi#:~:text=July%202020),%22courtyard%20shelter%22%20in%20English.)

⁵ *Study of libraries in python and their usage* <https://www.geeksforgeeks.org/libraries-in-python/>

worked with NumPy to handle complex computations. While Numpy allows sorting and indexing of array data, the numerical data code is stored in SciPy. It is also widely used by application developers and engineers.

Dataset:

The dataset used in this research project was sourced from Kaggle, an online platform with datasets that are usable for machine learning. The title of the dataset is “Rural Health Statistics - India - 2005, 2019, 2020”, last updated in 2020. The contributor of the dataset, Vineeth Akkinapalli, has claimed that this data set is obtained from the report from the Ministry of Health and Family Welfare, Government of India, 2019-2020. The dataset consisted of 9 columns namely State, PHC(Primary Healthcare Centre), CHC(Community Health Centers), ANM(Auxiliary Nursing Midwifery), Doctors, Specialists, Radiographers, Pharmacists, Nursing Staff and Lab Technicians. Of these 9 columns, PHC(Primary Healthcare Centre), CHC(Community Health Centers), ANM(Auxiliary Nursing Midwifery), Doctors, Pharmacists and Nursing Staff are columns of DataFrame that are used to train the model. Target column is used for specifying the dependent variable, Sub Centre is the target column for this model.

Method:

When training the model it was beneficial to convert Pandas Dataframe into NumPy arrays. Pandas is excellent for data manipulation and analysis, with its DataFrame structure allowing for easy handling of tabular data. However, when it comes to mathematical computations, NumPy arrays are more efficient due to their static typing and contiguous memory storage.⁶ The latter seems to be more memory efficient, especially when it comes to performing some complex mathematical operations over the data. The code converts the selected and the target columns DataFrame and the target column to NumPy arrays.⁷

Next, we split the data set into train and test sets to prepare it for the next step for both models. The train set is the part of the data that we worked with to create the algorithm, and the test set is what we tested the algorithm on.⁸ It is essential to split the model into training and test dataset as the evaluation process of the model needs to be done based on test data. That is an instance where a machine learning model fits its training data too well and fails to reliably fit additional data.⁹ We can understand it as if we train our model with a training set and then test it with a completely different test dataset, and then our

⁶ *Difference between data frame and numpy arrays* <https://towardsdatascience.com/pandas-df-to-numpy-array-c1a9e7d8585f>

⁷ *Better alternative between data frame and numpy array* <https://saturncloud.io/blog/convert-pandas-dataframe-to-numpy-array-which-a-approach-to-prefer/>

⁸ *Difference between test and train dataset in machine learning* <https://www.javatpoint.com/train-and-test-datasets-in-machine-learning>

⁹ *Data splitting and its need in machine learning* <https://www.techtarget.com/searchenterpriseai/definition/data-splitting#:~:text=In%20machine%20learning%2C%20data%20splitting,into%20three%20or%20four%20sets>

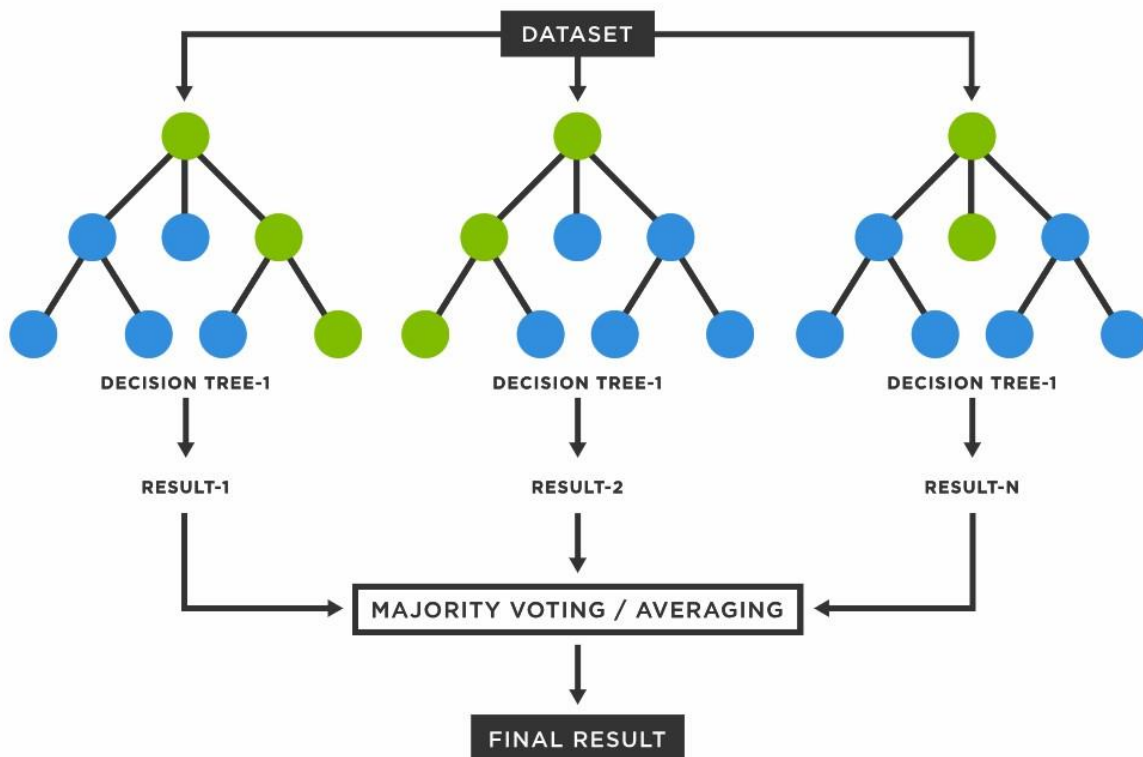
model will not be able to understand the correlations between the features. The model was split into a training dataset of 67 percent and testing dataset of 33 percent.

The model trains using three models: Random Forest, Linear Regression, and Neural Network regression.¹⁰

Model:

Random Forest Classifier

Random Forest is a classifier that contains a number of decision trees on various columns of the given dataset and takes the average to improve the predictive accuracy of that dataset. Instead of relying on one decision tree, the random forest takes the prediction from each tree and based on the majority votes of predictions, and it predicts the final output. The greater number of trees in the forest leads to higher accuracy and prevents the problem of overfitting.¹¹



¹⁰ Training a machine learning model on data <https://domino.ai/blog/what-is-machine-learning-model-training>

¹¹ .Study on Random forest algorithm <https://www.javatpoint.com/machine-learning-random-forest-algorithm>

Figure 1. Random Forest Classifier. Image Credit: TIBCO Software

Linear Regression

Linear regression analysis is used to predict the value of a variable based on the value of another variable. The goal of the algorithm is to find the best linear equation that can predict the target columns based on other columns of the dataset. The purpose of the model is to find weights of the equation in the training phase that can be used to predict the target column in the testing phase. The equation in the linear regression model had 6 variables as 6 columns were used to train the model.¹²

Neural Network Model

Regression analysis using artificial neural networks (ANNs) is a statistical technique used to predict the value of a dependent variable based on one or more independent variables. ANNs are trained on a dataset that includes selected and target values for a set of observations and can handle non-linear relationships and large amounts of data. However, they can be more difficult to interpret than other regression models and may require more data and computational resources to train.¹³

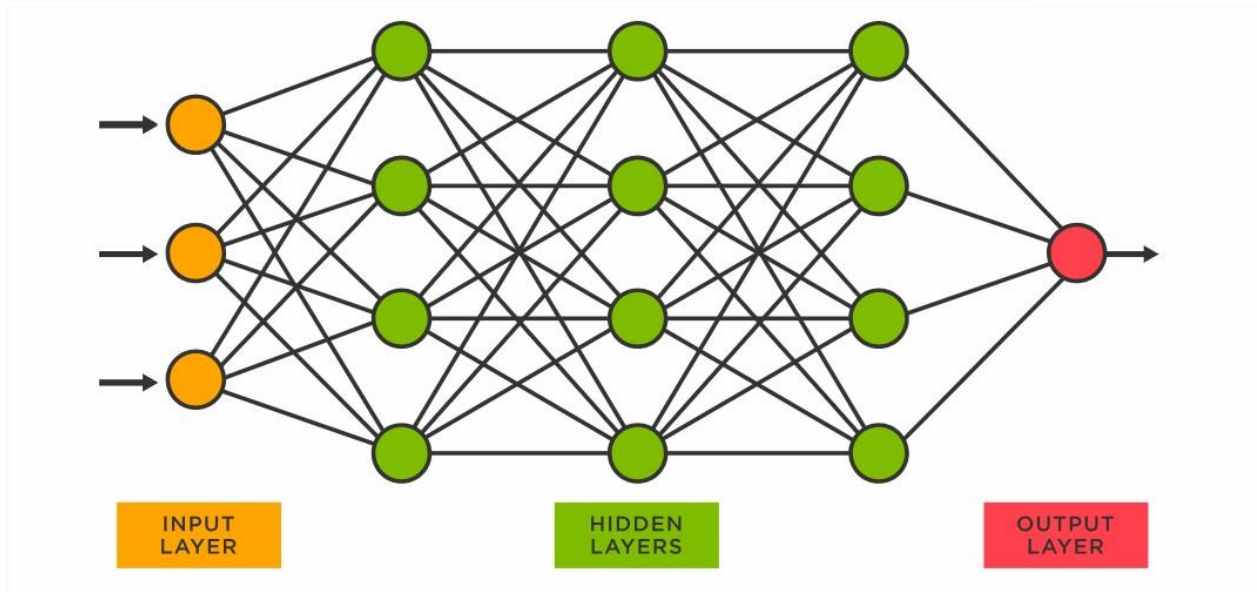


Figure 2. Neural Networking. Image Credit: TIBCO Software

Evaluation:

¹² Study of linear regression model <https://www.geeksforgeeks.org/ml-linear-regression/>

¹³ Study of the neural network model <https://www.scaler.com/topics/deep-learning/multiple-linear-regression/>

Next, the trained models were put in an evaluation stage based on the testing data (33% of overall data).

¹⁴Model evaluation is the process of using different evaluation metrics to understand a machine learning model's performance, as well as its strengths and weaknesses. Mean squared error (MSE) was used to measure the amount of error in the statistical models. It assessed the average squared difference between the observed and predicted values.

Results:

Following are the mean squared error results-

	Linear Regression Model	Random Forest Classifier	Neural Network
Testing data	14.4%	17%	50k
Training data	0.005	0.0097	2.53
Percentage Difference	0.04	0.05	0.005

Table 1 Train and Test Set Mean squared error of the Logistic Regression, Neural Network Model and Random Forest Classifier.

Since all the data points in the dataset had very large values, all of them were divided by 5000 to have a smaller scale of mean squared error. ¹⁵This, eventually, resulted in a mean squared error of 14.4% for the

¹⁴ *Training the model in the machine learning using the fit method*

<https://saturncloud.io/blog/what-is-the-fit-method-in-pythons-scikitlearn/#:~:text=The%20fit%20method%20is%20a,the%20model%20to%20the%20data.>

¹⁵ *.Study of .Mean squared error* <https://statisticsbyjim.com/regression/mean-squared-error-mse/>

linear regression model, 17% for the random forest classifier and 50,000% for the neural network model. The percentage difference of the linear regression model for training and testing data was as low as 0.04% and for the random forest classifier it was 0.05%, while for the neural network the percentage difference was 0.005%.